

PowerShell for Intune and SCCM Administrators

Zero-to-Advanced Master SOP, Lab Guide, Script Library, and Interview Book

Purpose

Designed for a learner starting with zero PowerShell knowledge. The guide explains why PowerShell is used, how the language works, how Intune and SCCM administrators apply it, how scripts are deployed, how Microsoft Graph is used, how Remediations work, how to troubleshoot scripts, and how to design safe enterprise automation.

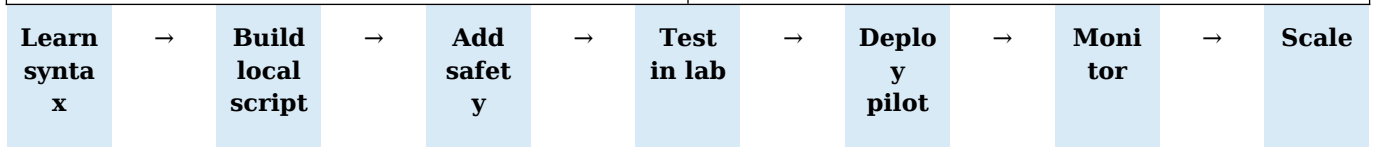
Document field	Value
Audience	Intune Administrator, Endpoint Engineer, SCCM/MECM Administrator, Intune SME candidate
Coverage	PowerShell language, Windows automation, Intune device/app/policy reporting, Intune scripts, Remediations, Microsoft Graph, SCCM cmdlets, packaging, troubleshooting
Starting level	Zero knowledge
Target level	Enterprise administrator and interview-ready architecture understanding
Version	1.0
Prepared date	25 September 2026
Classification	Internal learning and interview preparation

Responsible administration

All tenant changes must use least privilege, peer review, pilot scope, logging, and rollback. The samples prefer read-only operations. High-impact remote actions such as wipe, retire, delete, or bulk assignment are explained conceptually but are not provided as ready-to-run bulk commands.

Learning Roadmap

Stage	Outcome
1. PowerShell basics	Read, understand, and safely run commands
2. Script engineering	Build functions, logs, error handling, and reusable modules
3. Local endpoint automation	Inspect and configure Windows safely
4. Intune script delivery	Deploy Platform scripts and Remediations
5. Microsoft Graph	Read Intune data and perform controlled automation
6. Intune admin use cases	Devices, apps, policies, compliance, reports, Autopilot, security
7. SCCM automation	Use ConfigurationManager cmdlets and client/server automation
8. Enterprise design	Secrets, CI/CD, review, rollback, monitoring, interview preparation



1. PowerShell, Shells, and Versions

Why an Intune/SCCM administrator uses it

PowerShell provides an object-based automation language for Windows, Microsoft Graph, Intune, and Configuration Manager tasks.

Concept from zero

A shell accepts commands. PowerShell passes structured objects through a pipeline instead of plain text. Windows PowerShell 5.1 and PowerShell 7 can coexist; some SCCM scenarios still depend on the ConfigurationManager module and environment support.

Process

1. Open PowerShell without elevation for learning.
2. Check version and edition.
3. Inspect execution environment.
4. Run only read-only commands first.

PowerShell example

```
$PSVersionTable  
Get-Host  
Get-ExecutionPolicy -List  
Get-Command Get-Process
```

Real-world example

Scenario

An engineer finds a script works in PowerShell 7 but an older module requires Windows PowerShell. The version table identifies the host and edition before troubleshooting the script.

Troubleshooting

- Check `$PSVersionTable.PSEdition`.
- Confirm module support and architecture.
- Do not globally weaken execution policy as a shortcut.

Interview-ready answer

PowerShell is object-based automation. I first confirm engine version, module support, execution context, and privileges because those directly affect script behavior.

2. Cmdlets, Help, and Discovery

Why an Intune/SCCM administrator uses it

Administrators must find commands and syntax without memorizing every cmdlet.

Concept from zero

Cmdlets use Verb-Noun names. Get-Help explains syntax, parameters, examples, and notes. Get-Command discovers commands and modules. Get-Member reveals object properties and methods.

Process

5. Find the command.
6. Read current help.
7. Inspect examples.
8. Run a safe query.
9. Inspect output object.

PowerShell example

```
Update-Help # Run only from an approved admin workstation
Get-Command -Verb Get -Noun '*Device*'
Get-Help Get-Process -Full
Get-Process | Get-Member
```

Real-world example

Scenario

During interview preparation, the candidate does not remember an exact cmdlet. The candidate explains how Get-Command and Get-Help are used to discover the supported command and parameters.

Troubleshooting

- Online help may require internet.
- Module-specific commands appear only after module import.

Interview-ready answer

I do not rely on memory alone. I discover the command, read full help, inspect examples, and test a read-only query before using a write operation.

3. Variables, Data Types, and Operators

Why an Intune/SCCM administrator uses it

Intune scripts use variables for tenant IDs, device objects, paths, filters, results, and configuration values.

Concept from zero

Variables begin with \$. PowerShell infers types, but explicit types improve validation. Common types include string, integer, Boolean, datetime, array, hashtable, and custom object.

Process

10. Create variables.
11. Inspect type.
12. Use comparison and logical operators.
13. Avoid secrets in plain text.

PowerShell example

```
$DeviceName = 'WIN-PILOT-001'  
[int]$MaxAgeDays = 30  
[bool]$ExportEnabled = $true  
$Cutoff = (Get-Date).AddDays(-$MaxAgeDays)  
$DeviceName.GetType().FullName  
$MaxAgeDays -gt 0 -and $ExportEnabled
```

Real-world example

Scenario

A stale-device report uses an integer threshold and datetime cutoff rather than string comparison, preventing incorrect results.

Troubleshooting

- Use -eq, -ne, -gt, -lt, -like, -match correctly.
- Check null values before property use.

Interview-ready answer

Variables make scripts configurable, but I validate type and presence. Dates remain DateTime objects until final export or presentation.

4. Arrays, Hashtables, and Custom Objects

Why an Intune/SCCM administrator uses it

Reports and API payloads use collections and key-value data.

Concept from zero

Arrays store multiple values. Hashtables store named pairs and are useful for parameters, headers, and JSON payloads. PSCustomObject creates clean report rows.

Process

14. Create collection.
15. Loop or pipeline collection.
16. Build object with named properties.
17. Export structured result.

PowerShell example

```
$Platforms = @('Windows','macOS','iOS','Android')
$Row = [PSCustomObject]@{
    DeviceName = 'WIN-PILOT-001'
    Platform   = 'Windows'
    Compliant  = $true
    Collected = Get-Date
}
$Row | Format-List
```

Real-world example

Scenario

A report creates one custom object per managed device and exports a consistent schema to CSV.

Troubleshooting

- Avoid += on very large arrays; consider pipeline or generic lists.
- Keep property names stable.

Interview-ready answer

I use PSCustomObject for predictable reporting and hashtables for configuration or request bodies.

5. The Pipeline and Object Selection

Why an Intune/SCCM administrator uses it

PowerShell pipelines connect commands while preserving object properties.

Concept from zero

Where-Object filters objects, Select-Object chooses/creates properties, Sort-Object orders data, Group-Object summarizes, and Measure-Object counts or calculates.

Process

18. Get objects.
19. Filter as early as practical.
20. Select only required properties.
21. Sort or group.
22. Export without formatting cmdlets.

PowerShell example

```
Get-Service |  
Where-Object Status -eq 'Running' |  
Select-Object Name,Status,StartType |  
Sort-Object Name
```

Real-world example

Scenario

A device inventory filters Windows devices last seen before a cutoff and selects device name, OS, compliance, user, and last sync.

Troubleshooting

- Format-Table is for screen output, not CSV input.
- Use server-side filtering for Graph when supported.

Interview-ready answer

I treat each pipeline stage as Get, Filter, Shape, Sort/Group, and Export. I preserve objects until presentation.

6. Loops and Conditional Logic

Why an Intune/SCCM administrator uses it

Automation needs decisions per device, app, or policy.

Concept from zero

if/elseif/else makes decisions. foreach processes collections. switch handles multiple known values. Continue skips an item and break exits a loop.

Process

23. Validate condition.
24. Process one item.
25. Write structured result.
26. Handle unsupported state.

PowerShell example

```
$Devices = @('WIN-01','MAC-01')
foreach ($Device in $Devices) {
    if ($Device -like 'WIN-*) {
        [PSCustomObject]@{ Device=$Device; Action='Windows validation' }
    } else {
        [PSCustomObject]@{ Device=$Device; Action='Platform review' }
    }
}
```

Real-world example

Scenario

A remediation evaluates Windows only and returns NotApplicable for other platforms instead of failing.

Troubleshooting

- Prevent endless loops.
- Avoid write actions inside loops until dry-run evidence is reviewed.

Interview-ready answer

I validate scope before a loop and collect a proposed-action report before enabling changes.

7. Functions and Parameters

Why an Intune/SCCM administrator uses it

Functions turn repeated logic into tested, reusable commands.

Concept from zero

Advanced functions use CmdletBinding, typed parameters, validation, pipeline input, comment-based help, and ShouldProcess for changes.

Process

27. Define one responsibility.
28. Add parameters and validation.
29. Return objects, not formatted text.
30. Add error handling and WhatIf for changes.

PowerShell example

```
function Get-DeviceAgeState {  
    [CmdletBinding()]  
    param(  
        [Parameter(Mandatory)] [datetime]$LastSync,  
        [ValidateRange(1,3650)] [int]$ThresholdDays = 30  
    )  
    $Age = [math]::Floor(((Get-Date) - $LastSync).TotalDays)  
    [PSCustomObject]@{ AgeDays=$Age; IsStale=($Age -ge $ThresholdDays) }  
}  
Get-DeviceAgeState -LastSync (Get-Date).AddDays(-45)
```

Real-world example

Scenario

The same stale-age function is used in Intune and SCCM inventory reports with different thresholds.

Troubleshooting

- Keep functions small and testable.
- Use approved verbs from Get-Verb.

Interview-ready answer

Production scripts are collections of small functions with validated parameters and object output, not one long uncontrolled block.

8. Error Handling and Exit Codes

Why an Intune/SCCM administrator uses it

Intune and SCCM need clear success, detection, failure, and remediation status.

Concept from zero

try/catch/finally handles terminating errors. ErrorAction Stop converts many nonterminating errors. \$LASTEXITCODE applies to native executables. exit codes are especially important for Remediations and installers.

Process

31. Set strict error behavior deliberately.
32. Wrap risky operation in try/catch.
33. Log identity and context.
34. Return or exit with documented status.

PowerShell example

```
try {  
    $svc = Get-Service -Name 'IntuneManagementExtension' -ErrorAction Stop  
    [PSCustomObject]@{ Status='Success'; Service=$svc.Name; State=$svc.Status }  
}  
catch {  
    [PSCustomObject]@{ Status='Error'; Message=$_.Exception.Message }  
}
```

Real-world example

Scenario

A detection script exits 1 only when the issue exists; unexpected exceptions are logged distinctly rather than being treated as a compliant state.

Troubleshooting

- Do not suppress all errors with SilentlyContinue.
- Capture InvocationInfo and inner exception when needed.

Interview-ready answer

I separate expected noncompliance from script failure and use explicit exit codes and structured logging.

9. Logging and Transcripts

Why an Intune/SCCM administrator uses it

Logs provide auditability and troubleshooting evidence.

Concept from zero

A useful log contains timestamp, severity, script version, device/user context, action, result, and error. Transcripts capture console streams but may include sensitive data.

Process

35. Create approved log folder.
36. Use one logging function.
37. Mask secrets and personal data.
38. Rotate/retain according to policy.

PowerShell example

```
function Write-Log {  
    param([string]$Message,[ValidateSet('INFO','WARN','ERROR')]$Level='INFO')  
    "{0:o} [{1}] {2}" -f (Get-Date),$Level,$Message |  
        Add-Content -Path "$env:ProgramData\Company\Logs\IntuneAdmin.log"  
}  
Write-Log -Message 'Validation started'
```

Real-world example

Scenario

A Win32 app writes installer and wrapper logs to ProgramData so SYSTEM-context failures can be diagnosed consistently.

Troubleshooting

- Confirm SYSTEM can write the path.
- Never log tokens, passwords, recovery keys, or full sensitive payloads.

Interview-ready answer

Every enterprise script has versioned, privacy-safe logs that answer what ran, where, when, with which result.

10. Execution Policy, Script Signing, and Trust

Why an Intune/SCCM administrator uses it

Script-control choices affect security and supportability.

Concept from zero

Execution policy helps prevent accidental execution but is not a complete security boundary. Code signing, trusted repositories, App Control, constrained administration, and review are stronger controls.

Process

39. Keep enterprise execution settings approved.
40. Sign production scripts where policy requires.
41. Verify signature before deployment.
42. Store source in version control.
43. Review dependencies.

PowerShell example

```
Get-ExecutionPolicy -List
Get-AuthenticodeSignature -FilePath .\Script.ps1
# Set-AuthenticodeSignature requires an approved code-signing certificate.
```

Real-world example

Scenario

A downloaded script is blocked or untrusted. Instead of bypassing controls permanently, validate source, review code, unblock only if approved, and sign the maintained enterprise copy.

Troubleshooting

- Check Zone.Identifier, signature, and policy source.
- Do not use unrestricted policy as a permanent fix.

Interview-ready answer

I treat script trust as source control, peer review, signing, least privilege, and deployment governance, not just execution policy.

11. File, Registry, Service, and Process Basics

Why an Intune/SCCM administrator uses it

Many Intune scripts detect or remediate local Windows state.

Concept from zero

PowerShell providers expose filesystem, registry, certificate, and other stores. Services and processes are objects. Changes should be scoped and reversible.

Process

44. Query current state.
45. Compare to desired state.
46. Create backup or record.
47. Change only if required.
48. Re-query and verify.

PowerShell example

```
$Path = 'HKLM:\SOFTWARE\Company\Endpoint'  
Test-Path $Path  
Get-Service -Name 'WinDefend'  
Get-Process -Name 'OneDrive' -ErrorAction SilentlyContinue
```

Real-world example

Scenario

A detection script checks a registry value and a service state. It remediates only the missing value, then verifies.

Troubleshooting

- 32-bit and 64-bit registry views differ.
- User context and SYSTEM context see different HKCU profiles.

Interview-ready answer

Detection and remediation must be idempotent: query, compare, change minimally, and verify.

12. WMI, CIM, and System Inventory

Why an Intune/SCCM administrator uses it

CIM provides structured access to device hardware and operating-system data.

Concept from zero

Get-CimInstance uses CIM/WMI classes such as Win32_OperatingSystem, Win32_ComputerSystem, and Win32_BIOS. Remote use depends on network and authorization.

Process

49. Query local inventory.
50. Select stable identifiers.
51. Convert dates carefully.
52. Do not collect unnecessary data.

PowerShell example

```
$OS = Get-CimInstance Win32_OperatingSystem
$CS = Get-CimInstance Win32_ComputerSystem
$BIOS = Get-CimInstance Win32_BIOS
[PSCustomObject]@{
  ComputerName=$env:COMPUTERNAME
  OS=$OS.Caption
  Version=$OS.Version
  Manufacturer=$CS.Manufacturer
  Model=$CS.Model
  Serial=$BIOS.SerialNumber
}
```

Real-world example

Scenario

A remediation report includes model and OS to identify whether failure is concentrated on one hardware family.

Troubleshooting

- Some classes require elevation or return vendor-specific values.
- Normalize virtual-device serial values.

Interview-ready answer

I use CIM for local evidence and minimize data to the troubleshooting purpose.

13. JSON, CSV, and REST Fundamentals

Why an Intune/SCCM administrator uses it

Graph and modern APIs use JSON; administrators commonly export CSV for analysis.

Concept from zero

ConvertTo-Json serializes objects. ConvertFrom-Json parses JSON. Export-Csv preserves rows and columns. Invoke-RestMethod calls REST endpoints and returns objects.

Process

53. Build object.
54. Convert to JSON with adequate depth.
55. Send only to approved endpoint.
56. Parse response.
57. Export selected properties.

PowerShell example

```
$Body = [PSCustomObject]@{
  displayName = 'Pilot report'
  enabled = $true
} | ConvertTo-Json -Depth 5
$Body
$Rows = @([PSCustomObject]@{Name='Device1';State='Compliant'})
$Rows | Export-Csv .\Report.csv -NoTypeInfo -Encoding UTF8
```

Real-world example

Scenario

Graph request body fails because nested JSON is truncated at insufficient depth. Increase depth and inspect payload before sending.

Troubleshooting

- Never place secrets in exported files.
- Use UTF-8 and deliberate date format.

Interview-ready answer

I validate the PowerShell object first, then serialize to JSON, and log only non-sensitive request metadata.

14. Modules and Package Management

Why an Intune/SCCM administrator uses it

Install, update, import, and pin approved modules.

Concept from zero

Microsoft.Graph.Authentication and workload modules; ConfigurationManager module for SCCM.

Process

58. Use Get-Module -ListAvailable, Import-Module, Get-InstalledModule.
59. Validate scope and prerequisites.
60. Run read-only pilot.
61. Review output and logs.
62. Enable controlled change only after approval.
63. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Module not found on automation worker.

Troubleshooting

- Validate repository, version, dependencies, architecture, and offline strategy.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

15. Testing with Pester

Why an Intune/SCCM administrator uses it

Turn expected behavior into repeatable tests.

Concept from zero

Unit tests mock external calls; integration tests use lab resources.

Process

64. Test detection, parsing, filters, error handling, and WhatIf.
65. Validate scope and prerequisites.
66. Run read-only pilot.
67. Review output and logs.
68. Enable controlled change only after approval.
69. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

A script change breaks the compliant exit code.

Troubleshooting

- Run tests in source control before deployment.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

16. Script Repository and Version Control

Why an Intune/SCCM administrator uses it

Scripts need ownership, history, review, release, and rollback.

Concept from zero

Use Git branches, pull requests, tags, changelog, and semantic versioning.

Process

70. Separate source, tests, docs, configuration, and release artifact.
71. Validate scope and prerequisites.
72. Run read-only pilot.
73. Review output and logs.
74. Enable controlled change only after approval.
75. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

An emergency edit has no record.

Troubleshooting

- Require peer review and release notes.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

17. Local SYSTEM Versus User Context

Why an Intune/SCCM administrator uses it

Intune platform and app scripts may run as SYSTEM or signed-in user.

Concept from zero

SYSTEM has machine access but different profile, network, HKCU, and mapped drives.

Process

76. Test with matching context and architecture.
77. Validate scope and prerequisites.
78. Run read-only pilot.
79. Review output and logs.
80. Enable controlled change only after approval.
81. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Script works manually but fails from Intune.

Troubleshooting

- Remove user assumptions and use ProgramData for shared state.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

18. 32-bit Versus 64-bit PowerShell

Why an Intune/SCCM administrator uses it

Registry and filesystem redirection can change results.

Concept from zero

A 32-bit host sees redirected registry and system paths.

Process

82. Choose host architecture deliberately and test both.
83. Validate scope and prerequisites.
84. Run read-only pilot.
85. Review output and logs.
86. Enable controlled change only after approval.
87. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Detection reads wrong uninstall registry.

Troubleshooting

- Use explicit registry view and Intune 64-bit option where required.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

19. Intune Platform Scripts

Why an Intune/SCCM administrator uses it

Deploy one-time or recurring administrative scripts through Intune script management.

Concept from zero

Platform scripts use Intune Management Extension on supported Windows devices.

Process

88. Write idempotent script, select context/signature/64-bit settings, assign pilot, monitor.
89. Validate scope and prerequisites.
90. Run read-only pilot.
91. Review output and logs.
92. Enable controlled change only after approval.
93. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Script reports success but local state is unchanged.

Troubleshooting

- Review IME logs, exit behavior, context, and detection/verification.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

20. Intune Remediations Architecture

Why an Intune/SCCM administrator uses it

Detection finds the issue; remediation runs only for detected issue.

Concept from zero

Detection exit 0 means no issue; exit 1 triggers remediation. Output appears in reporting subject to limits.

Process

94. Design condition, safe fix, verification, schedule, pilot, monitor.
95. Validate scope and prerequisites.
96. Run read-only pilot.
97. Review output and logs.
98. Enable controlled change only after approval.
99. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Stale certificate is detected and user notified or corrected.

Troubleshooting

- Distinguish noncompliance from script error and make remediation idempotent.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

21. Remediation Detection Template

Why an Intune/SCCM administrator uses it

Build deterministic, fast, non-destructive detection.

Concept from zero

Return useful output and explicit exit.

Process

100. Query exact desired state and handle exceptions.
101. Validate scope and prerequisites.
102. Run read-only pilot.
103. Review output and logs.
104. Enable controlled change only after approval.
105. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Detection always returns 0 due to swallowed error.

Troubleshooting

- Use try/catch and tests for compliant, noncompliant, and error states.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

22. Remediation Fix Template

Why an Intune/SCCM administrator uses it

Repair only the detected issue and verify.

Concept from zero

The fix must be repeatable and safe if rerun.

Process

106. Record old state, change, query new state, return result.
107. Validate scope and prerequisites.
108. Run read-only pilot.
109. Review output and logs.
110. Enable controlled change only after approval.
111. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Service restart temporarily fixes but config remains wrong.

Troubleshooting

- Fix root configuration and validate persistence.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

23. Certificates Remediation

Why an Intune/SCCM administrator uses it

Detect expiry, wrong issuer, missing trust, or stale certificate.

Concept from zero

Use Cert: provider and approved issuer/thumbprint rules.

Process

112. Detect first; remove only exact approved stale certificate.
113. Validate scope and prerequisites.
114. Run read-only pilot.
115. Review output and logs.
116. Enable controlled change only after approval.
117. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Wi-Fi fails from expired user certificate.

Troubleshooting

- Protect private keys and avoid broad certificate deletion.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

24. Service Health Remediation

Why an Intune/SCCM administrator uses it

Check startup type, running state, and dependency.

Concept from zero

Services may be disabled by policy or third-party product.

Process

118. Detect desired state, change only approved service, verify.
119. Validate scope and prerequisites.
120. Run read-only pilot.
121. Review output and logs.
122. Enable controlled change only after approval.
123. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

IME service stopped after device issue.

Troubleshooting

- Review policy ownership before remediation.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

25. Disk Space and Cache Remediation

Why an Intune/SCCM administrator uses it

Low disk can break apps, updates, and enrollment.

Concept from zero

Measure system drive and approved cache paths.

Process

124. Detect threshold, remove only approved temporary content, report reclaimed space.
125. Validate scope and prerequisites.
126. Run read-only pilot.
127. Review output and logs.
128. Enable controlled change only after approval.
129. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Updates fail on low disk.

Troubleshooting

- Never delete user data or unknown caches.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

26. Registry Configuration Remediation

Why an Intune/SCCM administrator uses it

Registry-based settings are common but may be policy-owned.

Concept from zero

A remediation must not fight MDM/GPO.

Process

130. Identify authority, detect value, set only if approved, verify.
131. Validate scope and prerequisites.
132. Run read-only pilot.
133. Review output and logs.
134. Enable controlled change only after approval.
135. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Browser policy keeps reverting.

Troubleshooting

- Resolve policy source rather than repeatedly forcing registry.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

27. Application Presence and Version Detection

Why an Intune/SCCM administrator uses it

Detect app by MSI product code, file version, registry, service, or package evidence.

Concept from zero

Use the most stable vendor-supported evidence.

Process

136. Inspect clean install, upgrade, uninstall.
137. Validate scope and prerequisites.
138. Run read-only pilot.
139. Review output and logs.
140. Enable controlled change only after approval.
141. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

App exists but version detection fails.

Troubleshooting

- Handle x86/x64 and per-user installs.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

28. Intune Management Extension Troubleshooting

Why an Intune/SCCM administrator uses it

IME executes Win32 apps, scripts, and related workloads.

Concept from zero

Logs and agent state reveal policy, download, execution, and reporting.

Process

142. Check service, last sync, logs, context, content, command, detection.
143. Validate scope and prerequisites.
144. Run read-only pilot.
145. Review output and logs.
146. Enable controlled change only after approval.
147. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Win32 app stuck retrying.

Troubleshooting

- Correlate IME logs with installer logs and portal timestamps.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

29. Microsoft Graph Concepts

Why an Intune/SCCM administrator uses it

Graph exposes Intune devices, apps, policies, assignments, reports, and actions.

Concept from zero

Use delegated or application permissions and least privilege.

Process

148. Authenticate, call v1.0 when available, handle paging, errors, throttling, and audit.
149. Validate scope and prerequisites.
150. Run read-only pilot.
151. Review output and logs.
152. Enable controlled change only after approval.
153. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Portal export needs automation.

Troubleshooting

- Start read-only and validate official resource documentation.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

30. Graph PowerShell SDK Setup

Why an Intune/SCCM administrator uses it

The SDK wraps Graph authentication and requests.

Concept from zero

Install approved module and connect with required scopes.

Process

154. Use Microsoft.Graph.Authentication and Invoke-MgGraphRequest.
155. Validate scope and prerequisites.
156. Run read-only pilot.
157. Review output and logs.
158. Enable controlled change only after approval.
159. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Consent error occurs.

Troubleshooting

- Confirm permission type, admin consent, account, and tenant.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

31. Delegated Authentication

Why an Intune/SCCM administrator uses it

Interactive user identity authorizes the script.

Concept from zero

Best for admin-operated tools with MFA and least privilege.

Process

160. Connect with read scopes, inspect context, disconnect.
161. Validate scope and prerequisites.
162. Run read-only pilot.
163. Review output and logs.
164. Enable controlled change only after approval.
165. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Wrong tenant selected.

Troubleshooting

- Check Get-MgContext and tenant ID.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

32. Application Authentication

Why an Intune/SCCM administrator uses it

Unattended automation uses app identity, preferably certificate or managed identity.

Concept from zero

Avoid client secrets where possible.

Process

166. Register app, grant minimum application permissions, store key securely, monitor.
167. Validate scope and prerequisites.
168. Run read-only pilot.
169. Review output and logs.
170. Enable controlled change only after approval.
171. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Runbook token fails after secret expiry.

Troubleshooting

- Use certificate/managed identity and rotate credentials.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

33. Graph Permissions and Consent

Why an Intune/SCCM administrator uses it

Every request requires explicit permissions.

Concept from zero

Read.All differs from ReadWrite.All; delegated differs from application.

Process

172. Map script operations to minimum permission and document approval.
173. Validate scope and prerequisites.
174. Run read-only pilot.
175. Review output and logs.
176. Enable controlled change only after approval.
177. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

403 Forbidden from valid token.

Troubleshooting

- Inspect scopes/roles and resource permission.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

34. Graph Pagination

Why an Intune/SCCM administrator uses it

Large tenants return multiple pages.

Concept from zero

Follow @odata.nextLink until absent.

Process

178. Accumulate results or stream them safely.
179. Validate scope and prerequisites.
180. Run read-only pilot.
181. Review output and logs.
182. Enable controlled change only after approval.
183. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Report shows first 100 devices only.

Troubleshooting

- Always test count and pagination.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

35. Graph Filtering and Selection

Why an Intune/SCCM administrator uses it

Server-side \$filter and \$select reduce data and time when supported.

Concept from zero

OData support varies by resource.

Process

184. Encode URI and validate results.
185. Validate scope and prerequisites.
186. Run read-only pilot.
187. Review output and logs.
188. Enable controlled change only after approval.
189. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Filter returns unsupported query.

Troubleshooting

- Fall back to documented query or client-side filter.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

36. Graph Throttling and Retry

Why an Intune/SCCM administrator uses it

Services can return 429 or transient errors.

Concept from zero

Honor Retry-After and use exponential backoff.

Process

190. Retry bounded times and log request ID.
191. Validate scope and prerequisites.
192. Run read-only pilot.
193. Review output and logs.
194. Enable controlled change only after approval.
195. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Bulk report intermittently fails.

Troubleshooting

- Do not immediately hammer the API.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

37. Generic Graph Paging Function

Why an Intune/SCCM administrator uses it

Reusable function keeps API scripts consistent.

Concept from zero

Return all value pages and preserve metadata as needed.

Process

196. Use Invoke-MgGraphRequest and nextLink.
197. Validate scope and prerequisites.
198. Run read-only pilot.
199. Review output and logs.
200. Enable controlled change only after approval.
201. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Different endpoint returns no value collection.

Troubleshooting

- Validate response shape per resource.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

38. Managed Device Inventory

Why an Intune/SCCM administrator uses it

Admins report OS, version, ownership, compliance, user, and last sync.

Concept from zero

Use /deviceManagement/managedDevices read endpoint.

Process

202. Select only necessary fields and export securely.
203. Validate scope and prerequisites.
204. Run read-only pilot.
205. Review output and logs.
206. Enable controlled change only after approval.
207. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Asset tracker needs authoritative snapshot.

Troubleshooting

- Do not treat stale check-in as active state.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

39. Stale Device Reporting

Why an Intune/SCCM administrator uses it

Stale means no recent check-in according to an approved threshold, not automatically safe to delete.

Concept from zero

Compare LastSyncDateTime to cutoff and enrich ownership/user.

Process

208. Export proposed lifecycle list for review.
209. Validate scope and prerequisites.
210. Run read-only pilot.
211. Review output and logs.
212. Enable controlled change only after approval.
213. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Terminated-user device appears stale.

Troubleshooting

- Retire/delete only through approved lifecycle.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

40. Compliance Reporting

Why an Intune/SCCM administrator uses it

Summarize complianceState and investigate per-policy evidence.

Concept from zero

Overall state alone does not identify root cause.

Process

214. Group state, list noncompliant, then query detailed reports/policies.
215. Validate scope and prerequisites.
216. Run read-only pilot.
217. Review output and logs.
218. Enable controlled change only after approval.
219. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Dashboard count changes unexpectedly.

Troubleshooting

- Check time, filters, duplicates, and last sync.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

41. Device Ownership Reporting

Why an Intune/SCCM administrator uses it

Personal versus corporate ownership affects policy and inventory.

Concept from zero

Report, validate corporate identifiers, and correct only with approval.

Process

220. Generate exceptions and update through controlled workflow.
221. Validate scope and prerequisites.
222. Run read-only pilot.
223. Review output and logs.
224. Enable controlled change only after approval.
225. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Corporate laptop marked personal.

Troubleshooting

- Confirm serial and enrollment before change.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

42. Primary User Analysis

Why an Intune/SCCM administrator uses it

Primary user affects Company Portal and support context.

Concept from zero

Do not assume enrolledBy equals current primary user.

Process

226. Report device-user relationship and identify anomalies.
227. Validate scope and prerequisites.
228. Run read-only pilot.
229. Review output and logs.
230. Enable controlled change only after approval.
231. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Shared device has inappropriate primary user.

Troubleshooting

- Use approved relationship update process.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

43. Remote Sync Concepts

Why an Intune/SCCM administrator uses it

Sync requests a check-in; it does not guarantee instant policy completion.

Concept from zero

Remote actions require write permissions and device support.

Process

232. Target one device, log requester/reason, invoke, monitor.
233. Validate scope and prerequisites.
234. Run read-only pilot.
235. Review output and logs.
236. Enable controlled change only after approval.
237. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Policy pending after change.

Troubleshooting

- Check device online state and subsequent reporting.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

44. High-Impact Remote Actions

Why an Intune/SCCM administrator uses it

Wipe, retire, delete, fresh start, and reset passcode can disrupt users or remove data.

Concept from zero

PowerShell should add safeguards, approval, and exact targeting.

Process

238. Use two-person review, typed confirmation, WhatIf, and audit.
239. Validate scope and prerequisites.
240. Run read-only pilot.
241. Review output and logs.
242. Enable controlled change only after approval.
243. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Bulk action list contains wrong devices.

Troubleshooting

- No ready-to-run bulk destructive samples; use portal/change process.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

45. Application Inventory and Assignment Report

Why an Intune/SCCM administrator uses it

Graph can inventory mobile apps and assignment relationships.

Concept from zero

Different app types expose different properties.

Process

244. Query apps, normalize type/version/owner, query assignments, export.
245. Validate scope and prerequisites.
246. Run read-only pilot.
247. Review output and logs.
248. Enable controlled change only after approval.
249. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Unowned apps remain assigned.

Troubleshooting

- Handle paging and app-type-specific schema.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

46. Win32 App Failure Reporting

Why an Intune/SCCM administrator uses it

Combine app metadata and device/user install status when supported.

Concept from zero

Failure code needs context and installer/IME logs.

Process

250. Export failures by app, device, user, code, time.
251. Validate scope and prerequisites.
252. Run read-only pilot.
253. Review output and logs.
254. Enable controlled change only after approval.
255. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

New version causes widespread failure.

Troubleshooting

- Stop rollout through approved assignment change and validate root cause.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

47. Configuration Policy Inventory

Why an Intune/SCCM administrator uses it

Automate inventory of settings catalogs, device configurations, scope tags, assignments, owners, and versions.

Concept from zero

Different policy resources require normalized output.

Process

256. Collect metadata first; settings expansion second.
257. Validate scope and prerequisites.
258. Run read-only pilot.
259. Review output and logs.
260. Enable controlled change only after approval.
261. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Duplicate policies found.

Troubleshooting

- Do not auto-delete duplicates.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

48. Policy Assignment Audit

Why an Intune/SCCM administrator uses it

Find policies without assignments, broad assignments, exclusions, and overlaps.

Concept from zero

Assignment IDs must be resolved to group names carefully.

Process

262. Produce review report and risk flags.
263. Validate scope and prerequisites.
264. Run read-only pilot.
265. Review output and logs.
266. Enable controlled change only after approval.
267. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Pilot policy assigned to all users.

Troubleshooting

- Use peer review before correction.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

49. Compliance Policy Inventory

Why an Intune/SCCM administrator uses it

Report platform, settings count, assignments, actions, and grace.

Concept from zero

Use resource-specific official endpoints.

Process

268. Export metadata and assignment map.
269. Validate scope and prerequisites.
270. Run read-only pilot.
271. Review output and logs.
272. Enable controlled change only after approval.
273. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Device unsupported by assigned platform policy.

Troubleshooting

- Validate applicability and exceptions.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

50. Settings Catalog Reporting

Why an Intune/SCCM administrator uses it

Settings Catalog uses configuration policies and setting instances.

Concept from zero

Nested settings require careful parsing.

Process

274. Query policy, settings, assignments, normalize.
275. Validate scope and prerequisites.
276. Run read-only pilot.
277. Review output and logs.
278. Enable controlled change only after approval.
279. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Interview asks how to back up policy.

Troubleshooting

- Treat export as documentation, not guaranteed one-click restore.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

51. Endpoint Security Policy Reporting

Why an Intune/SCCM administrator uses it

Endpoint policies span antivirus, firewall, encryption, ASR, EDR, account protection, and baselines.

Concept from zero

Resources and templates differ.

Process

280. Build workload inventory and conflict ownership map.
281. Validate scope and prerequisites.
282. Run read-only pilot.
283. Review output and logs.
284. Enable controlled change only after approval.
285. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Same firewall setting in two policies.

Troubleshooting

- Use per-setting status before changing.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

52. Update Policy Reporting

Why an Intune/SCCM administrator uses it

Inventory update rings, feature policies, quality policies, and driver policies.

Concept from zero

Avoid mixing incompatible update APIs.

Process

286. Report assignments, deferrals, deadlines, target versions, status.
287. Validate scope and prerequisites.
288. Run read-only pilot.
289. Review output and logs.
290. Enable controlled change only after approval.
291. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Device belongs to two rings.

Troubleshooting

- Resolve assignment overlap first.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

53. Autopilot Device Reporting

Why an Intune/SCCM administrator uses it

Inventory serial, group tag, profile assignment, enrollment state, and linked managed device where available.

Concept from zero

Autopilot registration, Entra device, and Intune managed device are different records.

Process

292. Correlate stable IDs carefully.
293. Validate scope and prerequisites.
294. Run read-only pilot.
295. Review output and logs.
296. Enable controlled change only after approval.
297. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Registered device has no profile.

Troubleshooting

- Check group tag, dynamic membership, assignment, and sync.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

54. Group Membership Automation Concepts

Why an Intune/SCCM administrator uses it

Groups drive Intune assignment; unsafe membership automation creates blast radius.

Concept from zero

Use dynamic groups where appropriate or controlled Graph membership.

Process

298. Generate proposed add/remove list and require approval.
299. Validate scope and prerequisites.
300. Run read-only pilot.
301. Review output and logs.
302. Enable controlled change only after approval.
303. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Wrong devices enter update ring.

Troubleshooting

- Use stable identifiers and reconciliation report.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

55. Report Export Jobs

Why an Intune/SCCM administrator uses it

Intune reporting can use export jobs for supported reports.

Concept from zero

Report schema and filters vary; jobs are asynchronous.

Process

304. Submit job, poll state, download from returned location, validate.
305. Validate scope and prerequisites.
306. Run read-only pilot.
307. Review output and logs.
308. Enable controlled change only after approval.
309. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Portal report needs scheduled archive.

Troubleshooting

- Protect downloaded data and retention.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

56. Tenant Documentation and Backup

Why an Intune/SCCM administrator uses it

Automation can export metadata and JSON for review, diff, and recovery planning.

Concept from zero

Not every object has a simple portable restore path.

Process

310. Capture object type, ID, display name, JSON, assignments, dependencies, date.
311. Validate scope and prerequisites.
312. Run read-only pilot.
313. Review output and logs.
314. Enable controlled change only after approval.
315. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Policy deleted accidentally.

Troubleshooting

- Use recycle/restore where supported and approved recreation plan.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

57. Change Automation with ShouldProcess

Why an Intune/SCCM administrator uses it

Write operations should support dry run.

Concept from zero

CmdletBinding SupportsShouldProcess enables -WhatIf and -Confirm.

Process

316. Build proposed state, call ShouldProcess, log result.
317. Validate scope and prerequisites.
318. Run read-only pilot.
319. Review output and logs.
320. Enable controlled change only after approval.
321. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Assignment update targets broad group.

Troubleshooting

- Require approval and pilot even after WhatIf.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

58. Secrets, Certificates, and Managed Identity

Why an Intune/SCCM administrator uses it

Credentials must not be embedded in scripts or logs.

Concept from zero

Use managed identity, certificate authentication, or approved secret store.

Process

322. Grant least privilege and rotate/review.
323. Validate scope and prerequisites.
324. Run read-only pilot.
325. Review output and logs.
326. Enable controlled change only after approval.
327. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Script contains client secret in repository.

Troubleshooting

- Revoke, rotate, audit access, and remove history.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

59. Azure Automation and Scheduled Jobs

Why an Intune/SCCM administrator uses it

Scheduled reporting can run in Azure Automation or other governed platforms.

Concept from zero

Worker identity, modules, network, output, schedule, failure alert, and retention matter.

Process

328. Create read-only runbook first and monitor.
329. Validate scope and prerequisites.
330. Run read-only pilot.
331. Review output and logs.
332. Enable controlled change only after approval.
333. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Job succeeds but output missing.

Troubleshooting

- Review job streams, permissions, storage, and module version.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

60. CI/CD for Endpoint Scripts

Why an Intune/SCCM administrator uses it

Pipelines can lint, test, sign, package, and release scripts.

Concept from zero

Separate development from production and require approval.

Process

334. PSScriptAnalyzer, Pester, signing, artifact, release notes.
335. Validate scope and prerequisites.
336. Run read-only pilot.
337. Review output and logs.
338. Enable controlled change only after approval.
339. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Untested remediation deployed.

Troubleshooting

- Block release on failed tests.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

61. PowerShell Security Review

Why an Intune/SCCM administrator uses it

Review for dangerous operations, injection, secrets, excessive permissions, and weak validation.

Concept from zero

Treat external input as untrusted.

Process

340. Validate IDs, paths, URIs, and command arguments.
341. Validate scope and prerequisites.
342. Run read-only pilot.
343. Review output and logs.
344. Enable controlled change only after approval.
345. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

CSV input triggers wrong target.

Troubleshooting

- Use typed parameters and exact allowlists.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

62. SCCM Module and Site Drive

Why an Intune/SCCM administrator uses it

ConfigurationManager module connects PowerShell to the site.

Concept from zero

The site drive such as ABC: represents provider context.

Process

346. Import module, change site drive, run read-only Get-CMSite.
347. Validate scope and prerequisites.
348. Run read-only pilot.
349. Review output and logs.
350. Enable controlled change only after approval.
351. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Cmdlet fails from C: drive.

Troubleshooting

- Switch to site drive and verify provider.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

63. SCCM Device and Collection Queries

Why an Intune/SCCM administrator uses it

Automate inventory and collection review.

Concept from zero

Use Get-CMDevice and Get-CMCollection with supported parameters.

Process

352. Start read-only, avoid uncontrolled membership changes.
353. Validate scope and prerequisites.
354. Run read-only pilot.
355. Review output and logs.
356. Enable controlled change only after approval.
357. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Pilot collection contains unexpected devices.

Troubleshooting

- Inspect direct and query rules.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

64. SCCM Application Automation

Why an Intune/SCCM administrator uses it

Create/query applications and deployment types through the module.

Concept from zero

Application model depends on content, detection, requirements, distribution, deployment.

Process

358. Automate metadata and consistent naming carefully.
359. Validate scope and prerequisites.
360. Run read-only pilot.
361. Review output and logs.
362. Enable controlled change only after approval.
363. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Wrong detection causes false state.

Troubleshooting

- Validate in lab and monitor logs.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

65. SCCM Deployment Reporting

Why an Intune/SCCM administrator uses it

Use PowerShell and supported data sources to summarize deployment state.

Concept from zero

Cmdlet and reporting capabilities vary; SQL access must be governed.

Process

364. Prefer supported cmdlets/reports and avoid heavy database queries.
365. Validate scope and prerequisites.
366. Run read-only pilot.
367. Review output and logs.
368. Enable controlled change only after approval.
369. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Deployment status stale.

Troubleshooting

- Check state messages and summarization.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

66. SCCM Content Distribution

Why an Intune/SCCM administrator uses it

Automate distribution and status checks.

Concept from zero

Content must reach distribution points before clients install.

Process

370. Distribute to pilot DP/group, monitor, validate.
371. Validate scope and prerequisites.
372. Run read-only pilot.
373. Review output and logs.
374. Enable controlled change only after approval.
375. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Client gets content error.

Troubleshooting

- Review boundary, DP, distribution status, client logs.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

67. SCCM Software Update Automation

Why an Intune/SCCM administrator uses it

PowerShell can query updates, groups, deployments, and ADR objects through supported cmdlets.

Concept from zero

Patch automation carries high risk and change dependencies.

Process

376. Use pilot collections, deadlines, maintenance windows, and evidence.
377. Validate scope and prerequisites.
378. Run read-only pilot.
379. Review output and logs.
380. Enable controlled change only after approval.
381. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

ADR includes unwanted classification.

Troubleshooting

- Correct criteria and review update group.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

68. SCCM Client Automation

Why an Intune/SCCM administrator uses it

Local client actions can be triggered through approved mechanisms.

Concept from zero

Policy retrieval, inventory cycle, update scan, and state reporting are distinct.

Process

382. Use supported client methods and avoid repeated cycles.
383. Validate scope and prerequisites.
384. Run read-only pilot.
385. Review output and logs.
386. Enable controlled change only after approval.
387. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Client data stale.

Troubleshooting

- Check client health, WMI, policy, and logs.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

69. CMPivot and Run Scripts Concepts

Why an Intune/SCCM administrator uses it

SCCM can run approved scripts and real-time queries against collections.

Concept from zero

These are governance-controlled capabilities with audit implications.

Process

388. Sign/approve script, target pilot, monitor, retire.
389. Validate scope and prerequisites.
390. Run read-only pilot.
391. Review output and logs.
392. Enable controlled change only after approval.
393. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Script targets production collection.

Troubleshooting

- Confirm collection and approval before execution.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

70. Co-management Reporting

Why an Intune/SCCM administrator uses it

PowerShell can inventory enrollment and workload state.

Concept from zero

Workload ownership spans ConfigMgr and Intune.

Process

394. Report management authority, co-management state, policy source.
395. Validate scope and prerequisites.
396. Run read-only pilot.
397. Review output and logs.
398. Enable controlled change only after approval.
399. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Device receives duplicate policy.

Troubleshooting

- Trace workload slider, assignments, GPO, and Intune.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

71. Packaging Automation

Why an Intune/SCCM administrator uses it

PowerShell can prepare source, validate files, generate metadata, and call packaging tools.

Concept from zero

Automation should not invent silent switches.

Process

400. Read manifest, validate hash/version, package, generate logs.
401. Validate scope and prerequisites.
402. Run read-only pilot.
403. Review output and logs.
404. Enable controlled change only after approval.
405. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Wrong source version packaged.

Troubleshooting

- Verify source hash and detection.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

72. Installer Wrappers and Exit Codes

Why an Intune/SCCM administrator uses it

Wrappers standardize prerequisite checks, logging, installation, reboot, and detection.

Concept from zero

Native installer exit codes must be mapped deliberately.

Process

406. Capture process exit code and translate only documented values.
407. Validate scope and prerequisites.
408. Run read-only pilot.
409. Review output and logs.
410. Enable controlled change only after approval.
411. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

3010 treated as failure.

Troubleshooting

- Map vendor/MSI code with reboot behavior.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

73. Detection Script Engineering

Why an Intune/SCCM administrator uses it

Custom detection scripts should be fast, read-only, deterministic, and output minimal data.

Concept from zero

Exit 0 with output indicates detected for specific app scenarios; behavior depends on workload.

Process

412. Test present, absent, corrupt, and upgrade states.
413. Validate scope and prerequisites.
414. Run read-only pilot.
415. Review output and logs.
416. Enable controlled change only after approval.
417. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Per-user app under SYSTEM.

Troubleshooting

- Enumerate intended user context or choose device assignment carefully.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

74. Uninstall and Rollback Scripts

Why an Intune/SCCM administrator uses it

Every deployment needs an exit path.

Concept from zero

Uninstall must preserve user data according to requirements and verify removal.

Process

418. Test clean uninstall and rollback of configuration.
419. Validate scope and prerequisites.
420. Run read-only pilot.
421. Review output and logs.
422. Enable controlled change only after approval.
423. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Uninstall removes shared prerequisite.

Troubleshooting

- Model dependencies and shared components.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

75. Certificate and Network Diagnostics

Why an Intune/SCCM administrator uses it

PowerShell supports certificate store, DNS, routing, proxy, and TCP testing.

Concept from zero

Network issues often appear as Intune failures.

Process

424. Collect IP, DNS, proxy, TLS, certificate, and connectivity evidence.
425. Validate scope and prerequisites.
426. Run read-only pilot.
427. Review output and logs.
428. Enable controlled change only after approval.
429. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Enrollment fails behind proxy.

Troubleshooting

- Avoid changing enterprise proxy without approval.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

76. Defender and Firewall Diagnostics

Why an Intune/SCCM administrator uses it

Get-MpComputerStatus and firewall cmdlets expose local state.

Concept from zero

Local state must be correlated with policy ownership and portal reports.

Process

430. Query, export, compare expected state.
431. Validate scope and prerequisites.
432. Run read-only pilot.
433. Review output and logs.
434. Enable controlled change only after approval.
435. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Firewall rule succeeds but app fails.

Troubleshooting

- Check active store, path, profile, direction, protocol, address, port.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

77. BitLocker Validation

Why an Intune/SCCM administrator uses it

manage-bde and BitLocker cmdlets show encryption and protector state.

Concept from zero

Recovery keys are sensitive.

Process

436. Report encryption state without exposing key material.
437. Validate scope and prerequisites.
438. Run read-only pilot.
439. Review output and logs.
440. Enable controlled change only after approval.
441. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Compliance says unencrypted.

Troubleshooting

- Check policy, TPM, protection status, and escrow separately.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

78. Windows Update Diagnostics

Why an Intune/SCCM administrator uses it

PowerShell can collect update service state and generate Windows Update log.

Concept from zero

Do not force updates outside change policy.

Process

442. Collect ring/policy, service, reboot, and log evidence.
443. Validate scope and prerequisites.
444. Run read-only pilot.
445. Review output and logs.
446. Enable controlled change only after approval.
447. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Device stuck pending restart.

Troubleshooting

- Review deadlines, active hours, reboot state.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

79. Event Log Queries

Why an Intune/SCCM administrator uses it

Get-WinEvent supports targeted evidence collection.

Concept from zero

Use provider/log, time range, event ID, and correlation.

Process

448. Query narrow window and export selected fields.
449. Validate scope and prerequisites.
450. Run read-only pilot.
451. Review output and logs.
452. Enable controlled change only after approval.
453. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

MDM policy failed overnight.

Troubleshooting

- Use DeviceManagement Enterprise Diagnostics Provider log.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

80. Performance and Reliability

Why an Intune/SCCM administrator uses it

Large scripts need efficient queries, streaming, progress, timeouts, and cancellation.

Concept from zero

Measure before optimizing.

Process

454. Use server-side filters, selected properties, bounded concurrency.
455. Validate scope and prerequisites.
456. Run read-only pilot.
457. Review output and logs.
458. Enable controlled change only after approval.
459. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Tenant report times out.

Troubleshooting

- Checkpoint and retry transient failures.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

81. Data Privacy and Reporting

Why an Intune/SCCM administrator uses it

Intune data includes user, device, location, security, and identifiers.

Concept from zero

Collect minimum data and protect exports.

Process

460. Apply access, retention, encryption, and redaction.
461. Validate scope and prerequisites.
462. Run read-only pilot.
463. Review output and logs.
464. Enable controlled change only after approval.
465. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

CSV emailed broadly.

Troubleshooting

- Move to approved secure repository and restrict.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

82. Interview: Explain Your Script

Why an Intune/SCCM administrator uses it

Strong answers cover problem, data source, authentication, logic, safety, output, validation, and result.

Concept from zero

Do not start with syntax.

Process

466. Use architecture and evidence.
467. Validate scope and prerequisites.
468. Run read-only pilot.
469. Review output and logs.
470. Enable controlled change only after approval.
471. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Asked to write bulk wipe script.

Troubleshooting

- Explain safeguards and refuse unsafe unreviewed bulk action.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

83. Interview: Debug a Failed Script

Why an Intune/SCCM administrator uses it

Use scope, context, version, module, permission, input, API/log, and output layers.

Concept from zero

Reproduce in lab with same context.

Process

472. Isolate smallest failing step.
473. Validate scope and prerequisites.
474. Run read-only pilot.
475. Review output and logs.
476. Enable controlled change only after approval.
477. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

403 from Graph.

Troubleshooting

- Check token scopes/roles and endpoint permission.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

84. Interview: Design at Scale

Why an Intune/SCCM administrator uses it

Scale requires pagination, throttling, checkpoints, pilot, audit, and observability.

Concept from zero

Avoid serial one-device changes without design.

Process

478. Batch safely and produce proposed-action report.
479. Validate scope and prerequisites.
480. Run read-only pilot.
481. Review output and logs.
482. Enable controlled change only after approval.
483. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

5,000-device assignment change.

Troubleshooting

- Use staged rings and rollback.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

85. Capstone: Read-Only Intune Health Report

Why an Intune/SCCM administrator uses it

Combine device inventory, compliance, last sync, ownership, and OS summary.

Concept from zero

A capstone demonstrates Graph auth, paging, filtering, objects, export, logging, and privacy.

Process

484. Build functions, retrieve all pages, normalize, export, validate counts.
485. Validate scope and prerequisites.
486. Run read-only pilot.
487. Review output and logs.
488. Enable controlled change only after approval.
489. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Management requests weekly endpoint summary.

Troubleshooting

- Schedule only after manual validation.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

86. Capstone: Local Remediation Package

Why an Intune/SCCM administrator uses it

Create detection/remediation pair with logs and tests.

Concept from zero

Use a harmless setting such as approved service validation in lab.

Process

490. Write detect, remediate, verify, package, pilot, report.
491. Validate scope and prerequisites.
492. Run read-only pilot.
493. Review output and logs.
494. Enable controlled change only after approval.
495. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Issue returns after reboot.

Troubleshooting

- Fix persistent root setting.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

87. Capstone: SCCM Application Audit

Why an Intune/SCCM administrator uses it

Report application owner, deployment types, distribution and deployments.

Concept from zero

Use ConfigurationManager read-only cmdlets.

Process

496. Connect site, enumerate, normalize, export.
497. Validate scope and prerequisites.
498. Run read-only pilot.
499. Review output and logs.
500. Enable controlled change only after approval.
501. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status      = 'Review'
}
$Result
```

Real-world example

Scenario

Legacy apps have no owner.

Troubleshooting

- Use report for cleanup project, not automatic deletion.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

88. Final Production Checklist

Why an Intune/SCCM administrator uses it

Production automation needs ownership, requirements, permissions, tests, pilot, logs, rollback, privacy, and review.

Concept from zero

A script is a managed product.

Process

502. Complete design record and peer review.
503. Validate scope and prerequisites.
504. Run read-only pilot.
505. Review output and logs.
506. Enable controlled change only after approval.
507. Verify and document result.

PowerShell example

```
# Read-only discovery pattern
$Result = [PSCustomObject]@{
    ComputerName = $env:COMPUTERNAME
    CollectedUtc = (Get-Date).ToUniversalTime()
    Status       = 'Review'
}
$Result
```

Real-world example

Scenario

Quick admin script becomes business-critical.

Troubleshooting

- Version and support it formally.
- Compare working and failing samples.
- Record exact error, timestamp, context, module, and identity.

Interview-ready answer

I explain the requirement, management surface, script flow, safety controls, expected output, and troubleshooting evidence.

89. Practical Script Library: Graph Connection and Paging

These examples use placeholders and start with read-only scopes. Confirm current Microsoft Graph documentation and permissions before use.

```
# Install from an approved repository and workstation
# Install-Module Microsoft.Graph.Authentication -Scope CurrentUser
Import-Module Microsoft.Graph.Authentication

Connect-MgGraph -Scopes 'DeviceManagementManagedDevices.Read.All'
Get-MgContext | Select-Object TenantId,Account,Scopes

function Invoke-GraphPagedGet {
    [CmdletBinding()]
    param([Parameter(Mandatory)][string]$Uri)
    $All = [System.Collections.Generic.List[object]]::new()
    do {
        $Response = Invoke-MgGraphRequest -Method GET -Uri $Uri
        foreach($Item in @($Response.value)){ $All.Add($Item) }
        $Uri = $Response.'@odata.nextLink'
    } while($Uri)
    $All
}

$Uri = 'https://graph.microsoft.com/v1.0/deviceManagement/managedDevices?
$select=deviceName,operatingSystem,osVersion,complianceState,lastSyncDateTime,userPrincipalName,managedDeviceOwnerT
ype'
$Devices = Invoke-GraphPagedGet -Uri $Uri
$Devices.Count
Disconnect-MgGraph
```

Safety

The script reads device metadata. Store exports only in an approved secured location and minimize user fields.

90. Practical Script: Managed Device Inventory Report

```
$Cutoff = (Get-Date).ToUniversalTime().AddDays(-30)
$Report = $Devices | ForEach-Object {
    $LastSync = if($_.lastSyncDateTime){ [datetime]$_.lastSyncDateTime } else { $null }
    [PSCustomObject]@{
        DeviceName = $_.deviceName
        OS = $_.operatingSystem
        OSVersion = $_.osVersion
        Compliance = $_.complianceState
        Ownership = $_.managedDeviceOwnerType
        LastSyncUtc = $LastSync
        IsOlderThan30Days = ($LastSync -and $LastSync -lt $Cutoff)
        UserPrincipalName = $_.userPrincipalName
    }
}
$Report | Export-Csv '.\IntuneDeviceInventory.csv' -NoTypeInfo -Encoding UTF8
```

Operational note

The stale flag is for review, not automatic retirement or deletion.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

91. Practical Script: Compliance Summary

```
$Report |  
Group-Object Compliance |  
Sort-Object Count -Descending |  
Select-Object @ {n='ComplianceState';e={$_.Name}},Count |  
Export-Csv '.\ComplianceSummary.csv' -NoTypeInfo
```

Operational note

Follow overall state with per-policy investigation in approved reports.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

92. Practical Script: Safe Proposed-Action Report

```
$Candidates = $Report | Where-Object {  
    $_.IsOlderThan30Days -and $_.Compliance -ne 'compliant'  
}  
$Candidates | Select-Object DeviceName,OS,Compliance,LastSyncUtc,Ownership |  
Export-Csv '.\ProposedLifecycleReview.csv' -NoTypeInformation  
Write-Host 'No device action was taken. Review and approve the report.'
```

Operational note

Separating discovery from change is a critical control.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

93. Practical Script: Detection Script Template

```
try {  
    $Path = 'HKLM:\SOFTWARE\Company\Endpoint'  
    $Value = Get-ItemProperty -Path $Path -Name 'Configured' -ErrorAction Stop  
    if($Value.Configured -eq 1){  
        Write-Output 'Compliant: Configured=1'  
        exit 0  
    }  
    Write-Output 'Detected: Configured is not 1'  
    exit 1  
}  
catch {  
    Write-Output "Detected or unavailable: $($_.Exception.Message)"  
    exit 1  
}
```

Operational note

Customize the detection semantics to the intended workload and test errors separately.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

94. Practical Script: Remediation Script Template

```
$Path = 'HKLM:\SOFTWARE\Company\Endpoint'
try {
    if(-not (Test-Path $Path)){ New-Item -Path $Path -Force | Out-Null }
    New-ItemProperty -Path $Path -Name 'Configured' -Value 1 -PropertyType DWord -Force | Out-Null
    $Check = (Get-ItemProperty -Path $Path -Name 'Configured').Configured
    if($Check -ne 1){ throw 'Verification failed' }
    Write-Output 'Remediation successful'
    exit 0
}
catch {
    Write-Error $_.Exception.Message
    exit 1
}
```

Operational note

Use only in a lab under the correct authority; do not fight an existing MDM/GPO owner.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

95. Practical Script: Local Troubleshooting Snapshot

```
$Output = [ordered]@{}  
$Output.TimestampUtc = (Get-Date).ToUniversalTime()  
$Output.ComputerName = $env:COMPUTERNAME  
$Output.PowerShell = $PSVersionTable.PSVersion.ToString()  
$Output.Join = (dsregcmd /status | Select-String 'AzureAdJoined|DomainJoined|DeviceId') -join ';'   
$Output.IME = (Get-Service IntuneManagementExtension -ErrorAction SilentlyContinue).Status  
$Output.OS = (Get-CimInstance Win32_OperatingSystem).Caption  
[PSCustomObject]$Output | ConvertTo-Json -Depth 4 |  
Set-Content '.\EndpointSnapshot.json' -Encoding UTF8
```

Operational note

Review output for identifiers before sharing.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

96. Practical Script: SCCM Module Connection

```
# Run on a supported admin workstation with the console installed
Import-Module "$($env:SMS_ADMIN_UI_PATH)\..\ConfigurationManager.psd1"
$SiteCode = 'ABC'      # Replace with approved site code
Set-Location "$SiteCode\"
Get-CMSite
Get-CMDevice -CollectionName 'Pilot Devices' |
Select-Object Name,ClientType,ClientVersion,LastActiveTime
```

Operational note

Use a real site code only on the authorized Configuration Manager environment.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

97. Practical Script: ShouldProcess Change Function

```
function Set-ApprovedLocalSetting {  
    [CmdletBinding(SupportsShouldProcess,ConfirmImpact='High')]  
    param([Parameter(Mandatory)][ValidateSet(0,1)][int]$Value)  
    $Path='HKLM:\SOFTWARE\Company\Endpoint'  
    if($PSCmdlet.ShouldProcess($Path,"Set Configured=$Value")){  
        New-Item -Path $Path -Force | Out-Null  
        New-ItemProperty -Path $Path -Name Configured -Value $Value -PropertyType DWord -Force | Out-Null  
    }  
}  
Set-ApprovedLocalSetting -Value 1 -WhatIf
```

Operational note

WhatIf is necessary but not sufficient; still require peer review and pilot.

Validation

- Run in lab first.
- Review input and output.
- Check permissions and execution context.
- Verify no sensitive data is logged.
- Document rollback or state restoration.

98. Enterprise Script Review Checklist

Area	Questions
Purpose	What problem and approved owner?
Scope	Which tenant, group, devices, users, platform?
Permissions	Minimum Graph/Intune/SCCM role?
Identity	Delegated, app certificate, or managed identity?
Input	Typed, validated, and protected?
Logic	Idempotent and deterministic?
Safety	Read-only default, WhatIf, confirmation, pilot?
Output	Structured, privacy-safe, complete, paginated?
Errors	Try/catch, retries, request ID, exit codes?
Testing	Pester, lab, working/failing cases?
Operations	Logging, retention, monitoring, ownership?
Rollback	Documented and tested?

99. 12-Week Learning Plan

Week	Topic	Hands-on output
1	PowerShell shell, help, objects	20 read-only commands
2	Variables, pipeline, loops, functions	Inventory script
3	Errors, logs, tests, signing	Reusable script template
4	Windows services, registry, CIM, event logs	Endpoint snapshot
5	Intune Platform Scripts	Pilot script package
6	Remediations	Detection/remediation pair
7	Graph auth, permissions, paging	Read-only device report
8	Apps, policies, assignments reports	Tenant inventory
9	Compliance, updates, Autopilot	Health dashboard export
10	SCCM module and app/update queries	SCCM audit
11	CI/CD, secrets, automation	Reviewed release pipeline design
12	Capstone and mock interview	Portfolio and oral walkthrough

100. Interview Questions and Answer Framework

Why use PowerShell in Intune?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

Platform Script vs Remediation vs Graph automation?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do detection and remediation exit codes work?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you authenticate to Graph securely?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

Delegated vs application permissions?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you handle more than one page of Graph data?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you handle 429 throttling?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you prevent accidental bulk changes?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you troubleshoot a script that works manually but fails in Intune?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you protect credentials?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you validate Win32 detection?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How does PowerShell integrate with SCCM?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you design automation for 5,000 devices?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

How do you test and version scripts?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

What logs do you review?

Answer structure: business purpose, management surface, authentication/context, object/data flow, safety controls, implementation steps, validation evidence, troubleshooting, and a truthful real or design example.

101. Official References

- [Microsoft Learn: Intune devices and apps API overview](#)
- [Microsoft Learn: Working with Intune in Microsoft Graph](#)
- [Microsoft Learn: Use Remediations to detect and fix support issues](#)
- [Microsoft Learn: PowerShell scripts for Remediations](#)
- [Microsoft Learn: Configuration Manager PowerShell cmdlets](#)
- [Microsoft GitHub: mggraph-intune-samples](#)
- [Microsoft Intune automation and scripting samples](#)

Maintenance

Verify current cmdlet names, Graph resource versions, permissions, Intune navigation, licensing, and platform support before production use. Microsoft Graph beta resources can change and should not be selected merely because they expose more properties.